

COMP0171 Bayesian Deep Learning

2024 Past Paper — Model Answers

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1 Question 1: True/False [20 marks]

1(a) [2 marks]

Question

(T/F) Suppose there are two random variables x and y with joint distribution $p(x, y)$. If they have zero covariance, i.e. $\text{Cov}(x, y) = 0$, then they are independent, i.e. $p(x, y) = p(x)p(y)$.

Answer: False

Zero covariance 意味着 $\mathbb{E}[xy] = \mathbb{E}[x]\mathbb{E}[y]$, 但这只排除了线性依赖关系。两个变量可以有非线性依赖但 covariance 为零。

经典反例: 设 $x \sim \text{Uniform}(-1, 1)$, $y = x^2$ 。则 y 完全由 x 决定 (显然不 independent), 但由对称性:

$$\text{Cov}(x, y) = \mathbb{E}[xy] - \mathbb{E}[x]\mathbb{E}[y] = \mathbb{E}[x^3] - 0 = 0$$

因为 x^3 是奇函数, 在对称分布上的期望为零。所以

$$\text{Cov}(x, y) = 0 \not\Rightarrow \text{independence.}$$

注意: 对于 jointly Gaussian 的情况, zero covariance $\Leftrightarrow$ independence 是成立的, 但一般情况下不成立。

1(b) [2 marks]

Question

(T/F) Suppose we only know a functional form of a probability distribution up to a normalizing constant, e.g. for some $p(x|\theta) = \gamma(x, \theta)/Z$, where θ is a parameter of the distribution, the value of Z is unknown. Then it is not possible to perform maximum likelihood estimation for $p(x|\theta)$, maximizing with respect to θ , without first estimating Z .

Answer: False

MLE 对 θ 求解:

$$\theta_{\text{ML}} = \arg \max_{\theta} \sum_{i=1}^N \log p(x_i|\theta) = \arg \max_{\theta} \sum_{i=1}^N [\log \gamma(x_i, \theta) - \log Z(\theta)]$$

关键区分:

- 如果 Z 不依赖于 θ (即 Z 对 θ 是常数), 则 $\log Z$ 在优化中可忽略, 只需 $\gamma(x, \theta)$ 即可做 MLE。
- 即使 Z 依赖于 θ , 也有方法绕过直接计算 Z , 例如 score matching、contrastive divergence 等。

因此, "不可能做 MLE" 的表述过于绝对, 是错误的。

1(c) [2 marks]

Question

(T/F) If we are using self-normalized importance sampling to estimate an expectation $\mathbb{E}_{p(x)}[f(x)]$, where the normalizing constant of $p(x)$ is unknown, then for any number of samples and for any proposal $q(x)$ the resulting estimator using weights $w(x) \propto p(x)/q(x)$ is biased.

Answer: True

Self-normalized importance sampling estimator 的形式为:

$$\hat{\mu}_{\text{SNIS}} = \frac{\sum_{s=1}^S w(x^{(s)}) f(x^{(s)})}{\sum_{s=1}^S w(x^{(s)})}, \quad x^{(s)} \sim q(x)$$

这是一个 **ratio of two random variables** 的估计。由于

$$\mathbb{E} \left[\frac{A}{B} \right] \neq \frac{\mathbb{E}[A]}{\mathbb{E}[B]}$$

(Jensen's inequality), 对于任意有限的 sample 数 S , SNIS estimator 都是 biased 的。不过它是 asymptotically unbiased 的 ($S \rightarrow \infty$ 时 consistent), bias 为 $O(1/S)$ 。

1(d) [2 marks]

Question

(T/F) Reverse-mode autodiff can have higher memory requirements than forward-mode autodiff for functions composed of many sequential operations.

Answer: True

Reverse-mode autodiff (即 backpropagation) 需要先做一次 forward pass, 存储所有中间变量 (activations), 然后在 backward pass 中利用这些中间值计算梯度。对于由 L 个 sequential operations 组成的计算图, 需要存储 $O(L)$ 个中间结果。

相比之下, forward-mode autodiff 在 forward pass 中同时传播导数, 不需要存储所有中间变量, memory 需求更低。这正是深度网络训练中 memory 成为瓶颈的原因。Gradient checkpointing 等技术就是为了缓解这一问题。

1(e) [2 marks]

Question

(T/F) Adaptive learning rate approaches such as Adagrad and Adam avoid the need to manually tune any learning rates, instead setting them automatically during the optimization process.

Answer: **False**

Adagrad 和 Adam 会自适应调整每个参数的有效 learning rate, 但它们仍然有一个全局 **base learning rate** α (Adam 中通常默认为 10^{-3}) 需要手动设置。实践中这个 base learning rate 对训练效果有显著影响。

此外 Adam 还有 $\beta_1, \beta_2, \epsilon$ 等 hyperparameters 需要设定。虽然默认值通常 work 得不错, 但说“避免了手动调参的需要”是不准确的。

1(f) [2 marks]

Question

(T/F) Suppose there is a deep learning model for binary classification, with a maximum likelihood estimate θ^* and a predictive likelihood $p(y|x, \theta^*)$. If we use the Laplace approximation to estimate a posterior covariance matrix Σ and use the resulting predictive distribution that integrates over the posterior distribution $p(\theta|\theta^*, \Sigma)$, this will change predictive uncertainty but will not change the decision boundary.

Answer: **True**

Laplace approximation 以 θ^* (MLE 点) 为 posterior 均值:

$$p(\theta|\mathcal{D}) \approx \mathcal{N}(\theta|\theta^*, \Sigma), \quad \Sigma = (-\nabla_{\theta}^2 \log p(\theta|\mathcal{D})|_{\theta=\theta^*})^{-1}$$

Predictive distribution 变为:

$$p(y=1|x) = \int \sigma(\theta^{\top} \phi(x)) p(\theta|\theta^*, \Sigma) d\theta$$

由于 posterior 关于 θ^* 对称 (Gaussian), 对 logistic sigmoid 的积分在 $\theta^{*\top} \phi(x) = 0$ 处保持 $p(y=1|x) = 0.5$, 即 **decision boundary** 不变。直觉上, posterior 的对称性意味着 θ^* 两侧的贡献在 decision boundary 上完美平衡。

但在远离 decision boundary 的区域, marginalization 会使 predictive probabilities 向 0.5 靠拢, 因此 **uncertainty** 改变了, 但 **decision boundary** 没变。

1(g) [2 marks]

Question

(T/F) Markov chain Monte Carlo methods cannot easily be used to estimate the marginal likelihood.

Answer: **True**

MCMC 的设计目标是从 posterior $p(\theta|\mathcal{D})$ 中采样, 只需要 target distribution 的 unnormalized density $p(\mathcal{D}|\theta)p(\theta)$ 。Marginal likelihood (model evidence) 为:

$$p(\mathcal{D}) = \int p(\mathcal{D}|\theta)p(\theta) d\theta$$

MCMC samples 来自 posterior, 直接用它们估计这个积分需要额外的技巧 (如 harmonic mean estimator, 但 notoriously unstable), 或者 thermodynamic integration、nested sampling 等专门方法, 这些都不是标准 MCMC 的“轻松”应用。
相比之下, variational inference 通过 ELBO 天然提供 marginal likelihood 的下界, importance sampling 也可以直接估计。

1(h) [2 marks]

Question

(T/F) Linear regression, deep learning classification models, and Gaussian process regression all have likelihoods that factorize over datapoints, i.e. with $p(y_{1:N}|x_{1:N}) = \prod_{i=1}^N p(y_i|x_i)$, which allows for easy minibatching during training.

Answer: False

Linear regression 和 deep learning classification models (给定参数 θ) 的 likelihood 确实对 datapoints factorize:

$$p(y_{1:N}|x_{1:N}, \theta) = \prod_{i=1}^N p(y_i|x_i, \theta)$$

但 **Gaussian process regression** 不是这样。GP 的 marginal likelihood 为:

$$p(y_{1:N}|x_{1:N}) = \mathcal{N}(y_{1:N}|\mu, K + \sigma^2 I)$$

这是一个 **joint** Gaussian distribution, kernel matrix K 引入了所有 datapoints 之间的耦合。GP 不能简单地分解为各 datapoint 的独立贡献, 因此不能轻松做 minibatching (需要 inducing point 等近似方法)。

1(i) [2 marks]

Question

(T/F) Bi-directional recurrent neural networks can be used to extract features that depend on the entire length of a sequence, but cannot be used as a generative model.

Answer: True

Bi-directional RNN 在每个 time step 同时利用过去和未来的信息:

$$\vec{h}_t = f(\vec{h}_{t-1}, x_t), \quad \overleftarrow{h}_t = f(\overleftarrow{h}_{t+1}, x_t), \quad h_t = [\vec{h}_t; \overleftarrow{h}_t]$$

每个位置的 hidden state 编码了整个序列的信息, 适合做 feature extraction。

但 **generative model** 需要 **autoregressive** 分解: $p(x_1, \dots, x_T) = \prod_t p(x_t|x_{<t})$, 即生成 x_t 时只能依赖已生成的 $x_1, \dots, x_{t-1}$ 。Bi-directional RNN 在计算 h_t 时用了未来信息 $x_{t+1}, \dots, x_T$, 在生成阶段这些 token 尚不存在, 因此无法用于 autoregressive generation。

1(j) [2 marks]

Question

(T/F) In order for a Markov chain Monte Carlo method to correctly sample from a target distribution $p(x)$, the underlying Markov chain transition operator $T(x \rightarrow x')$ must satisfy the detailed balance condition $p(x)T(x \rightarrow x') = p(x')T(x' \rightarrow x)$.

Answer: **False**

正确采样要求 Markov chain 的 stationary distribution 为 $p(x)$, 即满足 **global balance**:

$$p(x') = \int p(x) T(x \rightarrow x') dx$$

Detailed balance 是一个充分条件 (sufficient condition), 不是必要条件 (necessary condition)。满足 detailed balance 一定满足 global balance (对 x 两边积分即得), 但反过来不成立。

存在不满足 detailed balance 但仍以 $p(x)$ 为 stationary distribution 的 MCMC 方法, 例如某些 non-reversible MCMC samplers (lifted MCMC、部分 Hamiltonian Monte Carlo 的 deterministic dynamics 等)。因此, "必须满足 detailed balance" 过强。

2 Question 2: Short Answer [30 marks]

2(a) [4 marks]

Question

Suppose we have data $\mathcal{D}$ and parameters θ , which for a given model class $\mathcal{M}$ have a joint distribution $p(\mathcal{D}, \theta | \mathcal{M})$. Briefly describe two approaches one could use to estimate the model evidence $p(\mathcal{D} | \mathcal{M}) = \int p(\mathcal{D}, \theta | \mathcal{M}) d\theta$.

Answer

方法一: Variational Inference

构造 variational distribution $q(\theta | \lambda)$, 通过最大化 ELBO 来近似 log marginal likelihood:

$$\log p(\mathcal{D} | \mathcal{M}) = \underbrace{\mathbb{E}_{q(\theta | \lambda)} \left[\log \frac{p(\mathcal{D}, \theta | \mathcal{M})}{q(\theta | \lambda)} \right]}_{\text{ELBO, } \mathcal{L}(\lambda)} + D_{\text{KL}}(q(\theta | \lambda) \| p(\theta | \mathcal{D}, \mathcal{M}))$$

由于 KL divergence ≥ 0 , $\mathcal{L}(\lambda^*) \leq \log p(\mathcal{D} | \mathcal{M})$, 可用作 model evidence 的近似下界。

方法二: Simple Monte Carlo / Importance Sampling

直接从 prior 中采样来估计:

$$p(\mathcal{D} | \mathcal{M}) = \int p(\mathcal{D} | \theta, \mathcal{M}) p(\theta | \mathcal{M}) d\theta = \mathbb{E}_{p(\theta | \mathcal{M})} [p(\mathcal{D} | \theta, \mathcal{M})]$$

$$\hat{p}(\mathcal{D} | \mathcal{M}) \approx \frac{1}{S} \sum_{s=1}^S p(\mathcal{D} | \theta^{(s)}, \mathcal{M}), \quad \theta^{(s)} \sim p(\theta | \mathcal{M})$$

这是 unbiased estimator, 但当 posterior 集中在 prior 的一个小区域时方差很大。可用 importance sampling (更好的 proposal $q(\theta)$) 或 Laplace approximation 改善。

2(b)(i) [3 marks]

Question

Suppose you have a deep learning model deployed on the camera for a self-driving car, which takes video of street scenes and segments it into labels, identifying the street, pavement, other cars, pedestrians, buildings, and any additional obstacles. This computer vision system has been trained on data from large amounts of data collected from city centres in the UK, both at daytime and nighttime.

What failure modes do you predict might arise for this system when deployed in the real world? Give one example where you would expect the model to have poor performance due to high **aleatoric uncertainty**.

Answer

Aleatoric uncertainty 是数据本身固有的噪声/模糊性, 即使有无限数据也无法消除。

例子: 大雾、暴雨或强烈反光条件下的场景。在这些条件下, 摄像头捕获的图像本身就是模糊的、信息不足的——路面和人行道边界因积水反射难以区分, 行人的轮廓被雨雾遮挡。这种不确定性来源于观测过程本身 (sensor noise), 而非模型知识不足。即使训练数据中包含此类场景, 单张模糊图像提供的信息仍不足以做出确定分割, uncertainty 是 irreducible 的。

2(b)(ii) [3 marks]

Question

Now give one example where you would expect the model to have poor performance due to high **epistemic uncertainty**.

Answer

Epistemic uncertainty 来源于模型对训练数据之外区域的知识不足, 可通过收集更多数据减少。

例子: 模型被部署到与训练环境截然不同的场景, 如日本或印度的街道。训练数据来自英国城市中心, 而日本有不同的交通标识、路面标记、建筑风格, 印度可能有三轮车、牛等英国不常见的物体。模型从未见过这些类别或视觉模式, uncertainty 高是因为训练数据覆盖不足 (out-of-distribution), 收集新场景标注数据后可减少。

2(c) [3 marks]

Question

What is a “deep ensemble”? Why could one consider this approach a form of approximate Bayesian inference in neural networks?

Answer

Deep ensemble 指独立训练 M 个具有相同架构但不同随机初始化的神经网络，预测时取平均：

$$p(y|x) \approx \frac{1}{M} \sum_{m=1}^M p(y|x, \theta_m)$$

为什么可视为 **approximate Bayesian inference**：在 Bayesian 框架中，predictive distribution 为：

$$p(y|x, \mathcal{D}) = \int p(y|x, \theta) p(\theta|\mathcal{D}) d\theta$$

Deep ensemble 可看作作用 M 个 point masses 近似 posterior： $p(\theta|\mathcal{D}) \approx \frac{1}{M} \sum_{m=1}^M \delta(\theta - \theta_m)$ 。由于 loss landscape 的多模态性，不同随机初始化收敛到不同 local optima，这些 θ_m 分布在 posterior 的不同 modes 上，从而捕获了 epistemic uncertainty。不过严格来说，ensemble 中每个 θ_m 是 MAP/MLE 解，并未真正按 posterior 采样。

2(d)(i) [4 marks]

Question

Suppose we are sampling from a probability distribution $p(x)$ using a Langevin MCMC algorithm. Given the gradient $\nabla_x \log p(x)$, write out the Langevin dynamics proposal distribution $q(x'|x)$ that would be used as part of a Metropolis-Hastings algorithm, and write out the MH acceptance ratio.

Answer

Langevin dynamics 的 proposal 为以梯度方向为 drift 的 Gaussian：

$$q(x'|x) = \mathcal{N}\left(x' \mid x + \eta \nabla_x \log p(x), 2\eta I\right)$$

等价地：

$$x' = x + \eta \nabla_x \log p(x) + \sqrt{2\eta} \epsilon, \quad \epsilon \sim \mathcal{N}(0, I)$$

其中 $\eta > 0$ 是 step size。这个 proposal 是非对称的（drift 项依赖于当前位置的梯度），所以 MH acceptance ratio 需要包含 proposal ratio：

$$\alpha = \min\left(1, \frac{p(x') q(x|x')}{p(x) q(x'|x)}\right)$$

其中 reverse proposal 为：

$$q(x|x') = \mathcal{N}\left(x \mid x' + \eta \nabla_{x'} \log p(x'), 2\eta I\right)$$

注意 forward 和 reverse proposal 的均值不同，分别依赖 $\nabla_x \log p(x)$ 和 $\nabla_{x'} \log p(x')$ 。

2(d)(ii) [4 marks]

Question

In class, we discussed a stochastic gradient Langevin dynamics algorithm which used a noisy estimate of the gradient, with no Metropolis-Hastings step. Why? Is this a valid MCMC sampler, and under what conditions?

Answer

SGLD (Welling & Teh, 2011) 每步更新为:

$$\theta^{(t+1)} = \theta^{(t)} + \eta_t \hat{g}(\theta^{(t)}) + \sqrt{2\eta_t} \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I)$$

其中 $\hat{g}$ 是 minibatch gradient estimate。

为什么跳过 MH step:

1. MH acceptance ratio 需要评估 full target density $p(\theta|\mathcal{D}) \propto p(\mathcal{D}|\theta)p(\theta)$ ，大数据下需遍历所有 N 个 datapoints，正是 SGLD 想避免的。
2. 数学上，当 $\eta \rightarrow 0$ 时，Langevin dynamics 的连续时间极限的 stationary distribution 就是 target distribution，acceptance probability $\rightarrow 1$ ，因此 MH correction 变得不必要。

是否 valid: 对于有限步长 $\eta > 0$ ，SGLD 不是精确的 MCMC sampler——stationary distribution 与 target 有偏差。但在以下条件下渐近精确:

- Step size η_t 按适当 schedule 递减至 0 (如满足 Robbins-Monro 条件 $\sum_t \eta_t = \infty$, $\sum_t \eta_t^2 < \infty$)
- 此时 gradient noise 被注入的 Gaussian noise $\sqrt{2\eta_t}\epsilon_t$ 所 dominate，SGLD 渐近地从 correct posterior 采样。

2(e) [6 marks]

Question

Convolutional layers have the “equivariance” property, that shifting an input will correspondingly shift the output. First, write out the equation for a one-dimensional convolutional layer, for a single length-3 filter $w \in \mathbb{R}^3$, as applied to an input $x \in \mathbb{R}^D$. (That is, write the expression for the d -th element of the one-dimensional convolution $h = \sigma(x * w)$, for a nonlinearity σ).

Second, show that the equivariant property holds by defining x' which is shifted to the right, i.e. with $x'_{d+1} = x_d$, would yield an output layer h' with $h'_{d+1} = h_d$.

[N.B. For purposes of this question you can ignore edge cases at x_1 and x_D .]

Answer

卷积层方程：对于 length-3 filter $w = (w_1, w_2, w_3)$ ，输出第 d 个元素为：

$$h_d = \sigma(w_1 x_{d-1} + w_2 x_d + w_3 x_{d+1})$$

其中 σ 是非线性激活函数，filter offset 为 $\{-1, 0, +1\}$ 。

证明 equivariance: 定义 shifted input x' : $x'_{d+1} = x_d$ (向右平移一位)，等价于 $x'_j = x_{j-1}$ 。则 x' 对应的输出为：

$$h'_d = \sigma(w_1 x'_{d-1} + w_2 x'_d + w_3 x'_{d+1})$$

利用 $x'_j = x_{j-1}$ 代入：

$$\begin{aligned} h'_d &= \sigma(w_1 x_{(d-1)-1} + w_2 x_{d-1} + w_3 x_{(d+1)-1}) \\ &= \sigma(w_1 x_{d-2} + w_2 x_{d-1} + w_3 x_d) \end{aligned} \quad (1)$$

而原始卷积中 d 替换为 $d-1$ ：

$$\begin{aligned} h_{d-1} &= \sigma(w_1 x_{(d-1)-1} + w_2 x_{d-1} + w_3 x_{(d-1)+1}) \\ &= \sigma(w_1 x_{d-2} + w_2 x_{d-1} + w_3 x_d) \end{aligned} \quad (2)$$

比较式 (1) 和 (2)： $h'_d = h_{d-1}$ ，即 $h'_{d+1} = h_d$ 。

输出也向右平移一位，equivariance 成立。 $\square$

2(f) [3 marks]

Question

Active learning and Bayesian optimization both involve iteratively collecting data and then updating a model. What is the difference, and when would prefer one approach over the other?

Answer

Active learning 的目标是学习一个好的 **predictive model**。它选择要标注的数据点使 model accuracy 提升最快 (acquisition 通常基于 model uncertainty)，最终目的是在标注预算有限时获得最好的泛化性能。

Bayesian optimization 的目标是找到 **black-box function** 的 **global optimum** $x^* = \arg \max_x f(x)$ 。它用 surrogate model (通常是 GP) 建模 f ，用 acquisition function (如 Expected Improvement, UCB) 决定下一个 evaluation point，目的是用尽可能少的 function evaluations 找到最优值。

选择标准：

- 有大量未标注数据、标注代价高、目标是整个 input space 上表现良好的模型 $\rightarrow$ **active learning**
- 评估代价高的 black-box function (物理实验、超参数调优)，只关心找最优点 $\rightarrow$

Bayesian optimization

3 Question 3: Bayesian Linear Classifiers [15 marks]

Setup: N observations $\{x_i, y_i\}$, $x_i \in \mathbb{R}^D$, $y_i \in \{0, 1\}$. Classifier: $\hat{y}_i = \sigma(\theta^\top \phi(x_i))$, σ is logistic sigmoid, $\theta \in \mathbb{R}^K$, $\phi: \mathbb{R}^D \rightarrow \mathbb{R}^K$ fixed features.

3(a) [3 marks]

Question

Suppose we would like to fit this using maximum likelihood estimation for θ , assuming the feature map ϕ is a fixed, known function. Write down an objective function you would optimize to find the maximum likelihood estimate $\hat{\theta}_{\text{ML}}$.

Answer

每个 datapoint 的 likelihood 为 Bernoulli:

$$p(y_i|x_i, \theta) = \hat{y}_i^{y_i} (1 - \hat{y}_i)^{1-y_i}, \quad \hat{y}_i = \sigma(\theta^\top \phi(x_i))$$

取 log 后 maximize (即 negative binary cross-entropy):

$$\hat{\theta}_{\text{ML}} = \arg \max_{\theta} \sum_{i=1}^N \left[y_i \log \sigma(\theta^\top \phi(x_i)) + (1 - y_i) \log (1 - \sigma(\theta^\top \phi(x_i))) \right]$$

3(b) [3 marks]

Question

Now suppose we would like to regularize our estimate of θ , by performing MAP estimation (maximum a posteriori) instead of maximum likelihood. To do this we include a Gaussian prior on θ , i.e. $\theta \sim \mathcal{N}(0, \beta I)$ where β is a prior variance. Write down a new objective function you would optimize to find the MAP estimate of $\hat{\theta}_{\text{MAP}}$.

Answer

MAP 估计 maximize posterior, 即 log-likelihood + log-prior:

$$\log p(\theta) = -\frac{K}{2} \log(2\pi\beta) - \frac{1}{2\beta} \|\theta\|^2$$

忽略与 θ 无关的常数:

$$\hat{\theta}_{\text{MAP}} = \arg \max_{\theta} \left[\sum_{i=1}^N \left(y_i \log \sigma(\theta^\top \phi(x_i)) + (1 - y_i) \log (1 - \sigma(\theta^\top \phi(x_i))) \right) - \frac{1}{2\beta} \|\theta\|^2 \right]$$

这等价于在 cross-entropy loss 上加 L2 regularization, regularization strength 为 $\lambda = \frac{1}{2\beta}$ 。

3(c) [3 marks]

Question

Assuming we are performing gradient descent to optimize θ , and suppose N is very large. In terms of the likelihood $p(y_i|x_i, \theta)$, define an estimator for the gradient of your objective in part (b) which uses only $M \ll N$ data points at each step, where the subset of M points is selected at random from the full dataset.

Answer

Part (b) 的 objective 的 full gradient 为:

$$\nabla_{\theta} \mathcal{L}(\theta) = \sum_{i=1}^N \nabla_{\theta} \log p(y_i|x_i, \theta) - \frac{1}{\beta} \theta$$

随机采样 minibatch $\mathcal{B} \subset \{1, \dots, N\}$, $|\mathcal{B}| = M$, stochastic gradient estimator 为:

$$\widehat{\nabla_{\theta} \mathcal{L}} = \frac{N}{M} \sum_{j \in \mathcal{B}} \nabla_{\theta} \log p(y_j|x_j, \theta) - \frac{1}{\beta} \theta$$

缩放因子 $\frac{N}{M}$ 保证 likelihood 部分是 full sum 的 unbiased estimate。Prior 项 $-\frac{1}{\beta} \theta$ 不依赖数据, 无需 subsampling。

3(d) [3 marks]

Question

Is your stochastic gradient estimator in part (c) an unbiased estimate of the true gradient? Show why or why not.

Answer

是 **unbiased** 的。设 $g_i(\theta) = \nabla_{\theta} \log p(y_i|x_i, \theta)$, full gradient 的 likelihood 部分为 $\sum_{i=1}^N g_i(\theta)$ 。

Minibatch estimator 为 $\hat{G} = \frac{N}{M} \sum_{j \in \mathcal{B}} g_j(\theta)$, 其中每个 j 是从 $\{1, \dots, N\}$ 中 uniform random 采的。

取期望:

$$\begin{aligned} \mathbb{E}_{\mathcal{B}}[\hat{G}] &= \frac{N}{M} \cdot M \cdot \mathbb{E}[g_j(\theta)] \\ &= \frac{N}{M} \cdot M \cdot \frac{1}{N} \sum_{i=1}^N g_i(\theta) \\ &= \sum_{i=1}^N g_i(\theta) \end{aligned}$$

这里利用了每个 j 是 uniform 从 $\{1, \dots, N\}$ 中采的, 所以 $\mathbb{E}[g_j] = \frac{1}{N} \sum_{i=1}^N g_i$ 。

加上 prior 项（确定性，无随机性）后， $\mathbb{E}[\text{stochastic gradient}] = \text{true gradient}$ ，即 **unbiased**。 □

3(e) [3 marks]

Question

Suppose you are now optimizing this objective via stochastic gradient descent on θ , and after 10 iterations where the loss reduces, on the 11th iteration the loss goes up. Should you now stop the optimization algorithm, or continue running gradient descent further? Explain your decision.

Answer

应该继续，不要停止。理由：

1. SGD 使用 stochastic gradients, loss 有随机波动是正常的。每次只用一个 minibatch 计算梯度，单次迭代 loss 上升完全符合预期（gradient noise 的自然结果），不意味着优化 diverge。
2. 判断收敛应看 loss 在多次迭代上的趋势（如 moving average），而非单步波动。
3. 仅 11 次迭代对模型训练远远不够。典型训练需要数百到数千次迭代。

正确做法是继续训练，监控 validation loss 的长期趋势，若 validation loss 持续上升多个 epoch 才考虑 early stopping。

4 Question 4: Gradient Estimators for Expectations [15 marks]

4(a) [5 marks]

Question

Suppose you wanted to estimate the derivative with respect to θ of a function $f(x, \theta) = x\theta^2$, in expectation under a normal distribution $p(x|\theta) = \mathcal{N}(x|\theta, 1)$, i.e. to estimate

$$\frac{d}{d\theta} \mathbb{E}_{p(x)}[f(x, \theta)].$$

Write down an unbiased Monte Carlo estimator for this derivative, using S samples. Be sure to specify all necessary details, including the function to evaluate as well as any sampling distributions.

Answer

θ 同时出现在 $f(x, \theta)$ 和 $p(x|\theta)$ 中。 $p(x|\theta) = \mathcal{N}(x|\theta, 1)$ 是连续分布且可 reparameterize，因此最佳选择是 **reparameterization trick**（pathwise gradient estimator），variance 通常比 score function estimator 低得多。

Reparameterization: 令 $\epsilon \sim \mathcal{N}(0, 1)$, 则 $x = \theta + \epsilon$ 。代入:

$$\mathbb{E}_{p(x|\theta)}[f(x, \theta)] = \mathbb{E}_{\mathcal{N}(\epsilon|0,1)}[(\theta + \epsilon)\theta^2]$$

对 θ 求导 (ϵ 的分布不依赖 θ , 导数可移入期望内):

$$\frac{d}{d\theta} \mathbb{E}_{p(\epsilon)}[(\theta + \epsilon)\theta^2] = \mathbb{E}_{p(\epsilon)} \left[\frac{d}{d\theta} [(\theta + \epsilon)\theta^2] \right]$$

计算:

$$\frac{d}{d\theta} [(\theta + \epsilon)\theta^2] = \frac{d}{d\theta} [\theta^3 + \epsilon\theta^2] = 3\theta^2 + 2\epsilon\theta$$

因此 unbiased Monte Carlo estimator 为:

$$\widehat{\frac{d}{d\theta} \mathbb{E}[f]} = \frac{1}{S} \sum_{s=1}^S (3\theta^2 + 2\epsilon^{(s)}\theta), \quad \epsilon^{(s)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$$

选择理由: $\mathcal{N}(x|\theta, 1)$ 有简单的 reparameterization $x = \theta + \epsilon$ 。Reparameterization trick 利用了 $\nabla_{\theta} f$ 的信息 (pathwise derivative), 而 score function estimator 不利用, 通常 variance 更高。

4(b) [5 marks]

Question

Suppose instead the distribution $p(x|\theta)$ was a geometric distribution, with

$$p(x = k|\theta) = \theta(1 - \theta)^{k-1}.$$

Repeat part (a), writing out an unbiased Monte Carlo estimator for the gradient of the expectation, for this new distribution.

Answer

Geometric distribution 是离散分布, 没有连续可微的 reparameterization。因此必须使用 **score function estimator** (REINFORCE)。

由于 θ 同时出现在 f 和 p 中, 完整的梯度为 (product rule applied to $\sum_k p(k|\theta)f(k, \theta)$):

$$\frac{d}{d\theta} \mathbb{E}_{p(x|\theta)}[f(x, \theta)] = \mathbb{E}_{p(x|\theta)} \left[f(x, \theta) \frac{d \log p(x|\theta)}{d\theta} + \frac{\partial f(x, \theta)}{\partial \theta} \right]$$

各项计算:

- $f(x, \theta) = x\theta^2$, 所以 $\frac{\partial f}{\partial \theta} = 2x\theta$
- Score function: $\log p(k|\theta) = \log \theta + (k-1) \log(1 - \theta)$

$$\frac{d}{d\theta} \log p(k|\theta) = \frac{1}{\theta} - \frac{k-1}{1-\theta}$$

因此 unbiased Monte Carlo estimator 为:

$$\widehat{\frac{d}{d\theta}\mathbb{E}[f]} = \frac{1}{S} \sum_{s=1}^S \left[x^{(s)} \theta^2 \left(\frac{1}{\theta} - \frac{x^{(s)} - 1}{1 - \theta} \right) + 2x^{(s)} \theta \right], \quad x^{(s)} \sim \text{Geom}(\theta)$$

4(c) [5 marks]

Question

Now suppose instead the distribution $p(x|\theta)$ is a discrete distribution that takes values $x \in \{1, 2, 3\}$ with probabilities $\theta_1, \theta_2, \theta_3$, respectively. Would you use the estimator above? If so, explain why. If not, write down an alternative computation for the gradient of the expectation.

Answer

不需要使用 Monte Carlo estimator。因为 x 只取 3 个值，期望可以精确计算（无需采样）：

$$\mathbb{E}_{p(x|\theta)}[f(x, \theta)] = \sum_{j=1}^3 \theta_j f(j, \theta) = \theta_1 f(1, \theta) + \theta_2 f(2, \theta) + \theta_3 f(3, \theta)$$

gradient 可以精确解析求解。对 θ_j 求偏导：

$$\frac{\partial}{\partial \theta_j} \mathbb{E}[f(x, \theta)] = f(j, \theta) + \sum_{k=1}^3 \theta_k \frac{\partial f(k, \theta)}{\partial \theta_j}$$

这是精确的解析表达式，无 variance。

为什么不用 score function estimator：当 support 有限且很小 ($|\mathcal{X}| = 3$) 时，精确求和的计算量可忽略不计，而 Monte Carlo estimator 会引入不必要的 variance。精确计算永远优于有 variance 的估计——Monte Carlo 只在精确求和不可行时（连续空间或离散空间太大）才有意义。

5 Question 5: Variational Inference over Weight Subsets [20 marks]

Setup: Regression problem, $y_i \sim \mathcal{N}(f(x_i, \theta), \sigma^2)$, f is a deep learning architecture with parameters θ .

5(a) [2 marks]

Question

Suppose we introduce a prior $p(\theta)$ over the parameters of the network, and wish to approximate the posterior using variational inference. If the approximating family has the form $q(\theta|\lambda)$, where λ are the variational parameters, write down the evidence lower bound

(ELBO), a lower bound for the log marginal likelihood $\sum_{i=1}^N \log p(y_i|x_i)$, that you would use as an objective function for learning λ .

Answer

$$\mathcal{L}(\lambda) = \mathbb{E}_{q(\theta|\lambda)} \left[\sum_{i=1}^N \log p(y_i|x_i, \theta) \right] - D_{\text{KL}}(q(\theta|\lambda) \| p(\theta))$$

等价写法:

$$\mathcal{L}(\lambda) = \mathbb{E}_{q(\theta|\lambda)} \left[\log \frac{p(\mathcal{D}|\theta)p(\theta)}{q(\theta|\lambda)} \right] \leq \log p(\mathcal{D})$$

其中 $p(y_i|x_i, \theta) = \mathcal{N}(y_i|f(x_i, \theta), \sigma^2)$ 。第一项是 expected log-likelihood, 第二项是 KL regularization, 保证 variational posterior 不偏离 prior 太远。

5(b) [4 marks]

Question

Now suppose we were to split the weights θ into two disjoint sets, θ_b and θ_p , where we aim to perform variational inference over θ_b and maximum a posteriori estimation for θ_p . Assuming that the prior $p(\theta)$ factorizes as $p(\theta) = p(\theta_b)p(\theta_p)$, write down an appropriate objective function for learning an approximate posterior $q(\theta_b|\lambda)$.

Answer

对 θ_b 做 VI, 对 θ_p 做 point estimation. Objective 为关于 λ 和 θ_p 的 joint objective:

$$\mathcal{L}(\lambda, \theta_p) = \mathbb{E}_{q(\theta_b|\lambda)} \left[\sum_{i=1}^N \log p(y_i|x_i, \theta_b, \theta_p) \right] - D_{\text{KL}}(q(\theta_b|\lambda) \| p(\theta_b)) + \log p(\theta_p)$$

各项解释:

- 第一项: expected log-likelihood, 在 $q(\theta_b|\lambda)$ 下取期望, θ_p 视为固定参数
- 第二项: KL divergence, regularize variational posterior 使其接近 prior $p(\theta_b)$
- 第三项: θ_p 的 log prior, 提供 MAP regularization

对 λ maximize 前两项 (标准 ELBO for θ_b); 对 θ_p maximize 第一项和第三项 (MAP estimation)。

5(c) [5 marks]

Question

Is the objective you defined above also appropriate for estimating θ_p ? Is it appropriate for estimating σ^2 ? Why or why not? Describe (high-level) the procedure you would use for estimating $q(\theta_b|\lambda)$, $\hat{\theta}_p$, and $\hat{\sigma}^2$.

Answer

对 θ_p : 是。同一个 objective $\mathcal{L}(\lambda, \theta_p)$ 可同时对 λ 和 θ_p 优化。对 θ_p 而言, maximize $\mathcal{L}$ 等价于 maximize expected log-likelihood + log prior, 正是对 θ_b 积分后的 MAP estimation。

对 σ^2 : 可以。 σ^2 作为 noise variance 出现在 likelihood $p(y_i|x_i, \theta) = \mathcal{N}(y_i|f(x_i, \theta), \sigma^2)$ 中。若不给 σ^2 加 prior, 可将它视为 hyperparameter, 通过 maximize ELBO (type-II ML / empirical Bayes) 估计。若加 prior 则做 MAP。两者都是对 $\mathcal{L}$ 关于 σ^2 求导。

整体 procedure (high-level):

联合优化或交替优化三组参数, 实际中通常合并为一个 stochastic gradient loop:

1. 每步从 $q(\theta_b|\lambda)$ 中采样 $\theta_b^{(s)}$ (用 reparameterization trick)
2. 计算 $\mathcal{L}$ 关于 λ 、 θ_p 、 σ^2 的 (stochastic) 梯度
3. 用 SGD/Adam 同时更新所有参数

对 likelihood 部分可做 minibatch subsampling 以 scale 到大数据集。

5(d) [5 marks]

Question

Given a new test input x_* , what is the form of the predictive distribution $p(y_*|x_*)$ using the model with an approximate posterior over θ_b and a point estimate for θ_p ? Write this as a (possibly intractable) integral or expectation, and describe how you would approximate it to make predictions.

Answer

Predictive distribution 需要对 θ_b 的 posterior 积分, 同时使用 θ_p 的 point estimate $\hat{\theta}_p$:

$$p(y_*|x_*) = \int p(y_*|x_*, \theta_b, \hat{\theta}_p) q(\theta_b|\lambda^*) d\theta_b = \mathbb{E}_{q(\theta_b|\lambda^*)} [\mathcal{N}(y_*|f(x_*, \theta_b, \hat{\theta}_p), \hat{\sigma}^2)]$$

这个积分通常 **intractable** (f 是非线性神经网络), 用 Monte Carlo 近似:

$$p(y_*|x_*) \approx \frac{1}{S} \sum_{s=1}^S \mathcal{N}(y_* | f(x_*, \theta_b^{(s)}, \hat{\theta}_p), \hat{\sigma}^2), \quad \theta_b^{(s)} \sim q(\theta_b|\lambda^*)$$

Predictive mean 和 variance:

$$\begin{aligned} \mathbb{E}[y_*|x_*] &\approx \frac{1}{S} \sum_{s=1}^S f(x_*, \theta_b^{(s)}, \hat{\theta}_p) \\ \text{Var}[y_*|x_*] &= \underbrace{\hat{\sigma}^2}_{\text{aleatoric}} + \underbrace{\text{Var}_{q(\theta_b)}[f(x_*, \theta_b, \hat{\theta}_p)]}_{\text{epistemic}} \\ &\approx \hat{\sigma}^2 + \frac{1}{S} \sum_{s=1}^S (f(x_*, \theta_b^{(s)}, \hat{\theta}_p) - \bar{f})^2 \end{aligned}$$

其中 $\bar{f} = \frac{1}{S} \sum_s f(x_*, \theta_b^{(s)}, \hat{\theta}_p)$ 。这里用了 law of total variance 将 total uncertainty 分解为 aleatoric 和 epistemic 两部分。

5(e) [4 marks]

Question

Give one reason you might want prefer the “Bayesian last layer” approach instead of the arbitrary splitting of weights described here, and one reason why you might prefer this approach instead.

Answer

偏好 **Bayesian last layer** 的理由:

当 last layer 是线性的 ($f(x, \theta) = \theta_{\text{last}}^\top h(x, \theta_{\text{rest}})$), 给定固定的 feature extractor h , posterior over θ_{last} 对应 Bayesian linear regression 问题。对 regression 且 Gaussian prior + Gaussian likelihood, posterior 有 **closed-form** 解 (不需要 variational inference), 计算效率极高且没有近似误差。相比之下, 对任意 weight subset 做 VI 需要优化 variational parameters, 有近似误差且计算更贵。

偏好 **arbitrary weight splitting** 的理由:

Bayesian last layer 只对最后一层 weights 量化 uncertainty, 可能低估 **epistemic uncertainty**。Feature extractor 的 weights 也有 uncertainty (尤其在 out-of-distribution 区域), 但被固定为 point estimate。允许对网络中更多层做 Bayesian inference 可以捕获更丰富的 uncertainty information, 对 safety-critical applications 的 uncertainty quantification 更有价值。